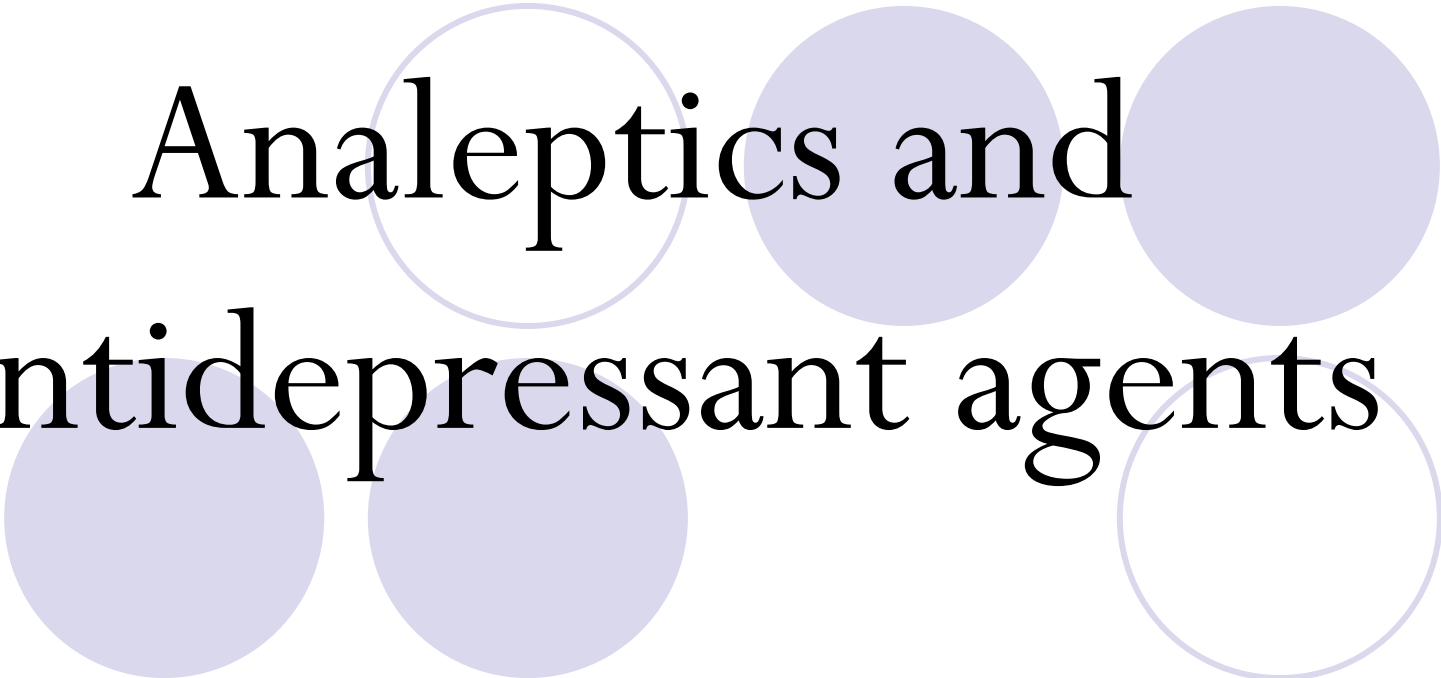




CNS Stimulants: Analeptics and Antidepressant agents

Yitagesu T. (MSc.)

Analeptics

Analeptic - A CNS stimulant that causes muscle contraction and perhaps also convulsions. Particularly a term used to describe compounds that cause contraction and rigidity of the muscles of respiration.

Strychnine is the most commonly recognized analeptic although it has relatively low potency.

Mechanisms of Action for CNS Stimulants:

1. Block neurotransmitter reuptake (Most reuptake inhibitors affect either NE or 5HT)

Tricyclic antidepressants

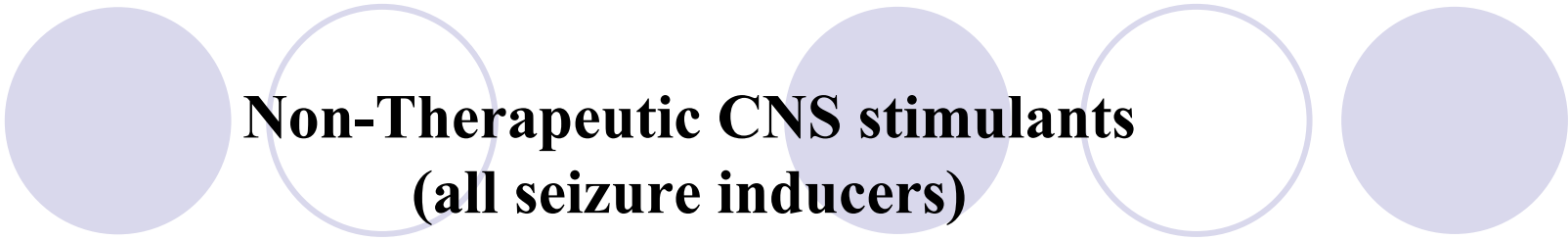
Cocaine

Selective Serotonin reuptake inhibitors

2. Promote Neurotransmitter Release

Phenylethylamines and related compounds. Amphetamine, Methylphenidate.

3. Block Metabolism - MAO inhibitors



Non-Therapeutic CNS stimulants (all seizure inducers)

Strychnine - Inhibits glycine receptors

Picrotoxin - Acts on the chloride ion channel associated with GABA receptors.

Bemigrade - Barbiturate Antagonist

Pentylenetetrazole - Na^+ / K^+ channel blocker
Increases cholinergic activity.

Xanthines (Caffeine, theophylline, theobromine)

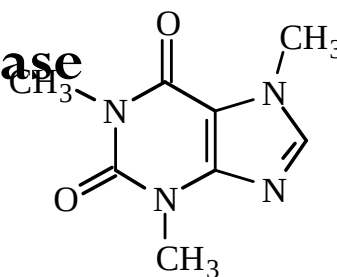
Mechanism(s) of action:

- Inhibition of **cAMP phosphodiesterase** through Competitive inhibitor of **adenosine** at adenosine A₁-receptor
- **Promote NE release**
- **Promote intracellular Ca^{+2} release**

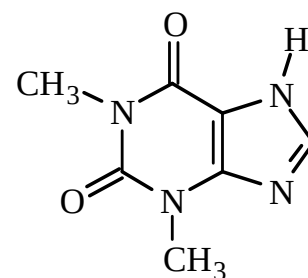
The order of potency for CNS activity is

Caffeine > Theophylline
> Theobromine

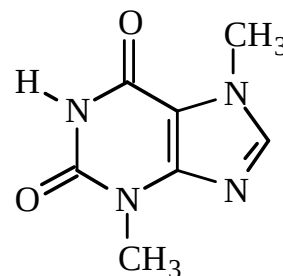
Theophylline is an important drug for maintenance treatment of asthma,



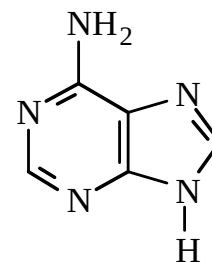
Caffeine



Theophylline



Theobromine

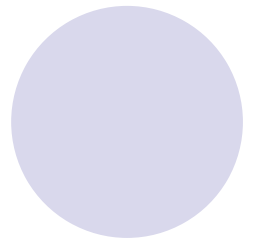
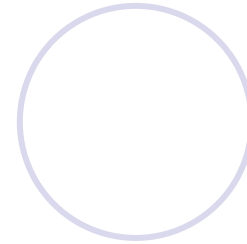
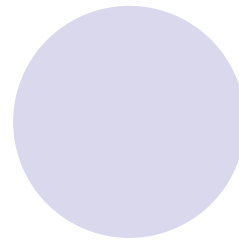
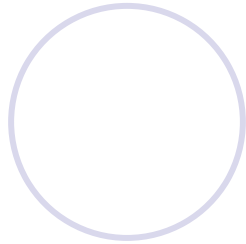
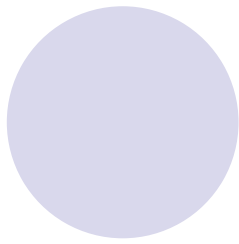


Adenine⁴



Phenylethylamines

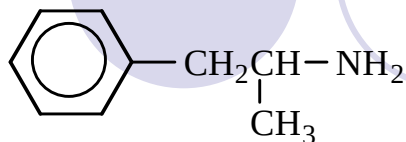
- Enhance neurotransmitter release
 - Block NT reuptake
 - Have direct agonist effects
 - MAO inhibition
-
- Phenylethylamines are used as **anorectics**, for **narcolepsy**, and not legitimate antidepressants.
 - The phenyl ring and the distance between the amine and the phenyl is fairly strict



With the exception of anorexia, many of the CNS effect of phenylethylamines are thought to involve effects on **dopamine** release and reuptake.

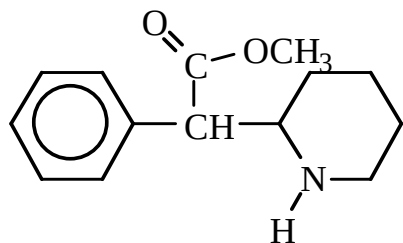
- “Amphetamine induces dopamine efflux through a dopamine transporter channel” PNAS (2005) vol. 102 | 3495-3500

Amphetamine Structures



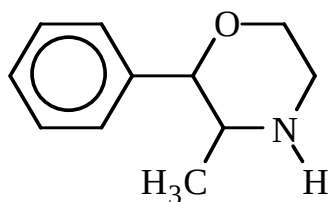
Amphetamine

Prototype phenylethylamine. The good, the bad, and the ugly

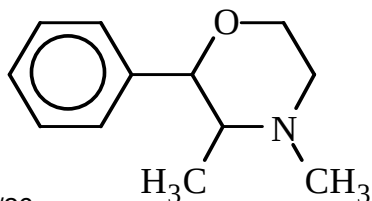


Methylphenidate
(Ritalin)

Used for attention-deficit-hyperactive disorder
Usually in children

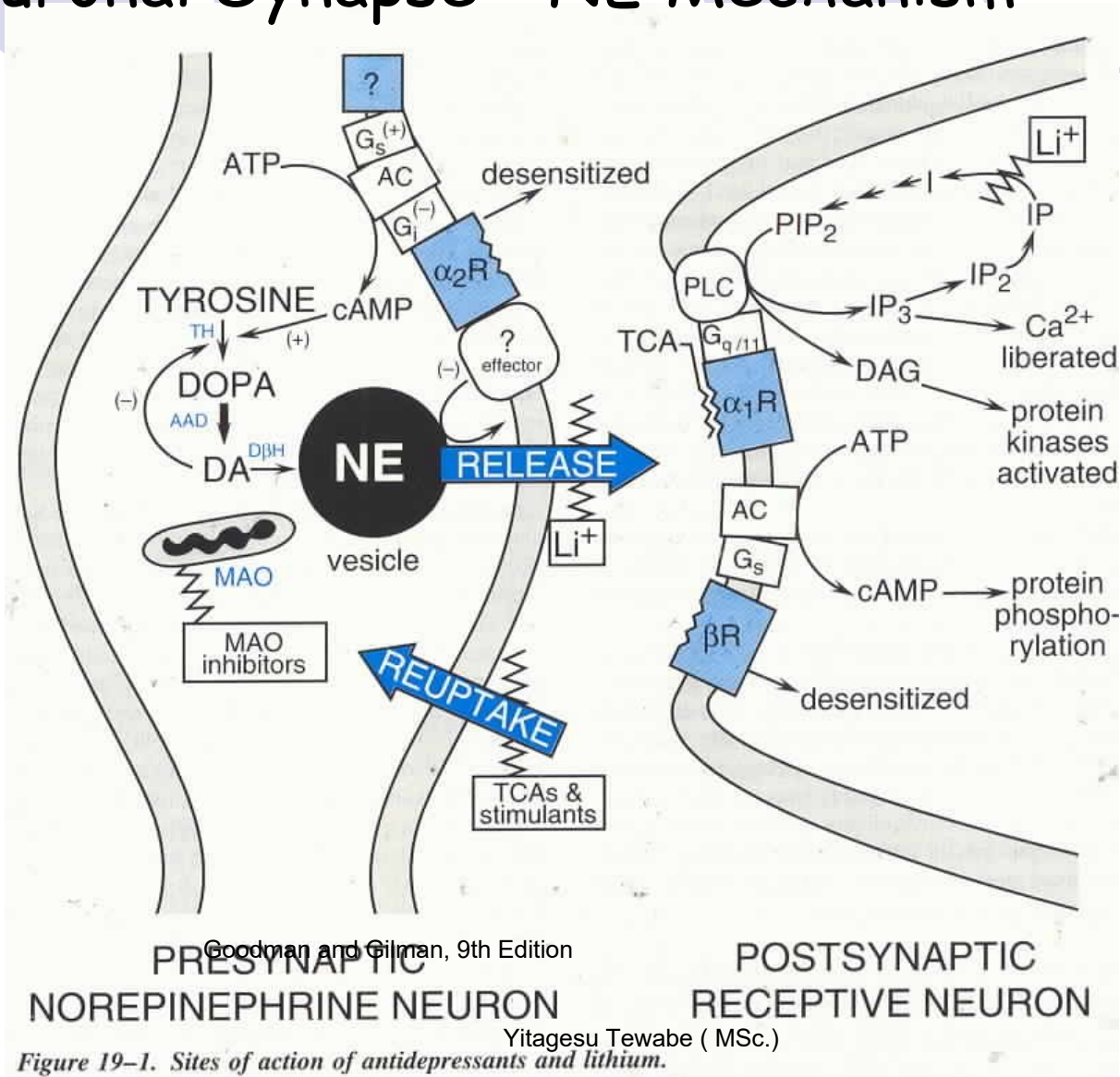


Phenmetrazine
(Preludin)
Anorectic

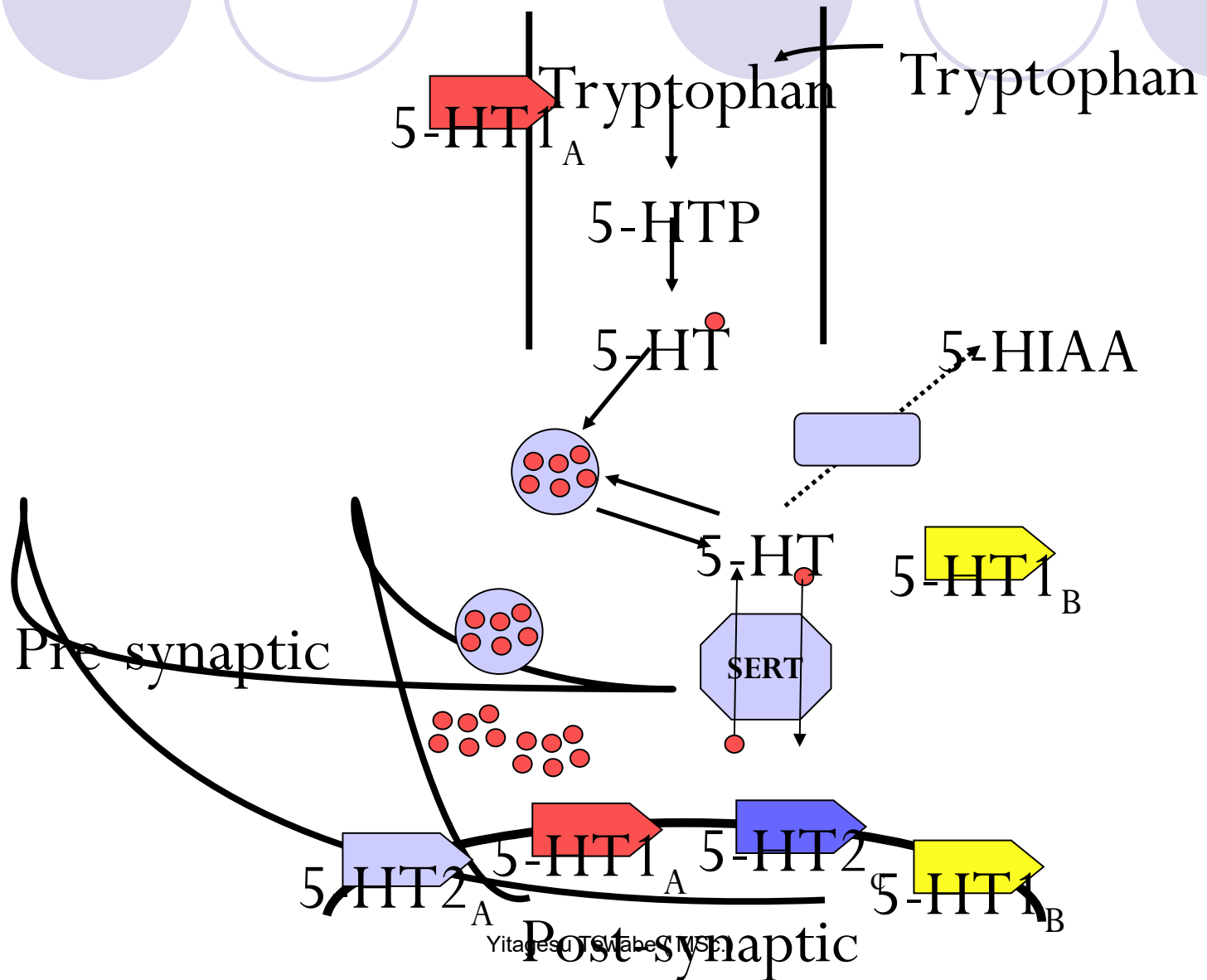


Phendimetrazine (Plegine)
Anorectic

Neuronal Synapse - NE Mechanism

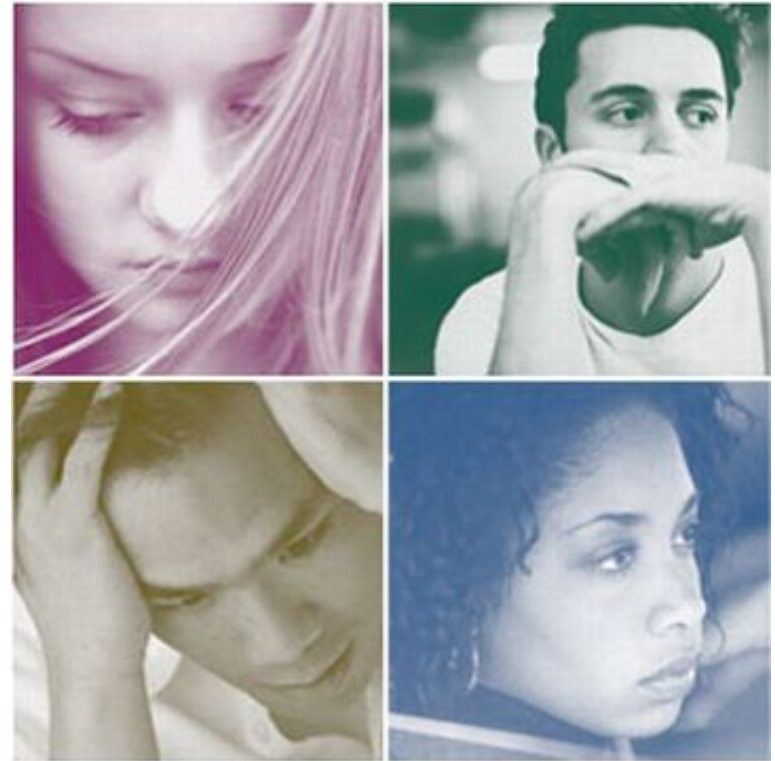


Neuronal Synapse - 5HT



Anti-depressants

- What is depression?
 - Types
 - Symptoms and Diagnosis
 - Biological Mechanism
- Antidepressants
 - History
 - Types (TCAs, MAOIs, SSRIs)
 - Mechanism of action
 - Side effects
 - Brain chemistry in the long-term



Definition of Depression



- **Neurological disorder**
- **Prolonged depression of mood and impairment of function**
- **Causes include**
 - genetic predisposition,
 - grief following the death of a loved one,
 - abuse,
 - major life changes, serious illness,
 - personal conflicts, and substance abuse
- **Complex illness with many contributing factors**
- **Exact biological causes are not yet fully understood**

Types

- Major
 - Symptoms interfere with ability to function normally
 - 2x more women are diagnosed than men
- Chronic (Dysthymia)
 - Less severe, but symptoms linger for longer
- Seasonal Affective Disorder (SAD)
- Psychotic
- Postpartum

Diagnosis and Symptoms

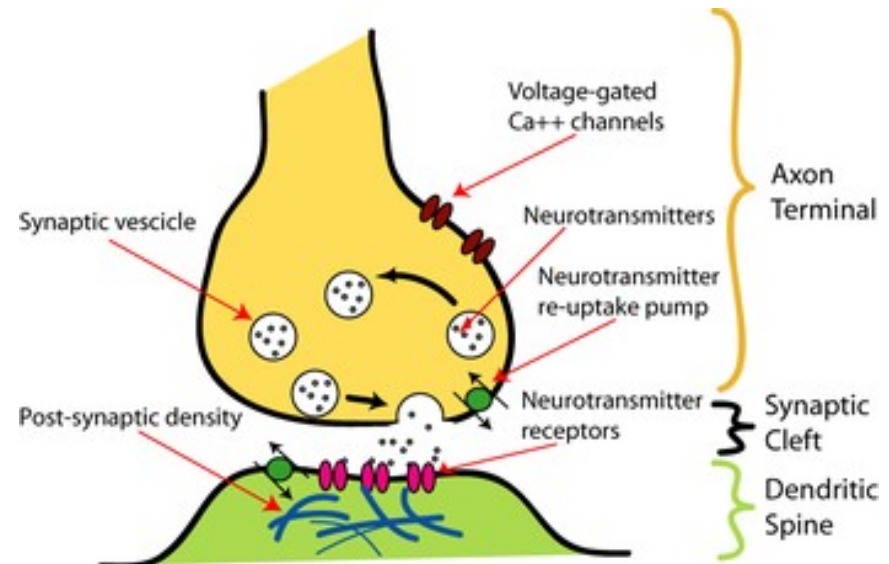
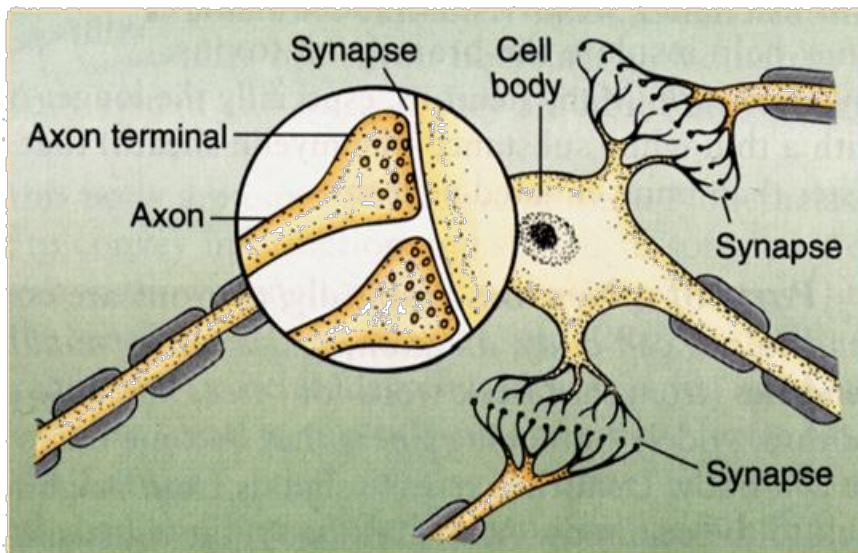
- See a doctor if symptoms persist for > 2 weeks
- Lab tests will be performed to rule out other diagnoses
- Psychometric tests follow
- Symptoms include:
 - Intense sadness or despair
 - Loss of concentration
 - Worry
 - Lack of pleasure
 - Self-deprecation
 - Agitation or hostility
 - Disruption in sleep patterns
 - Altered eating patterns
 - Lack of energy
 - Thoughts of suicide
 - Anxiety
 - Digestive problems



The neurochemistry of depression

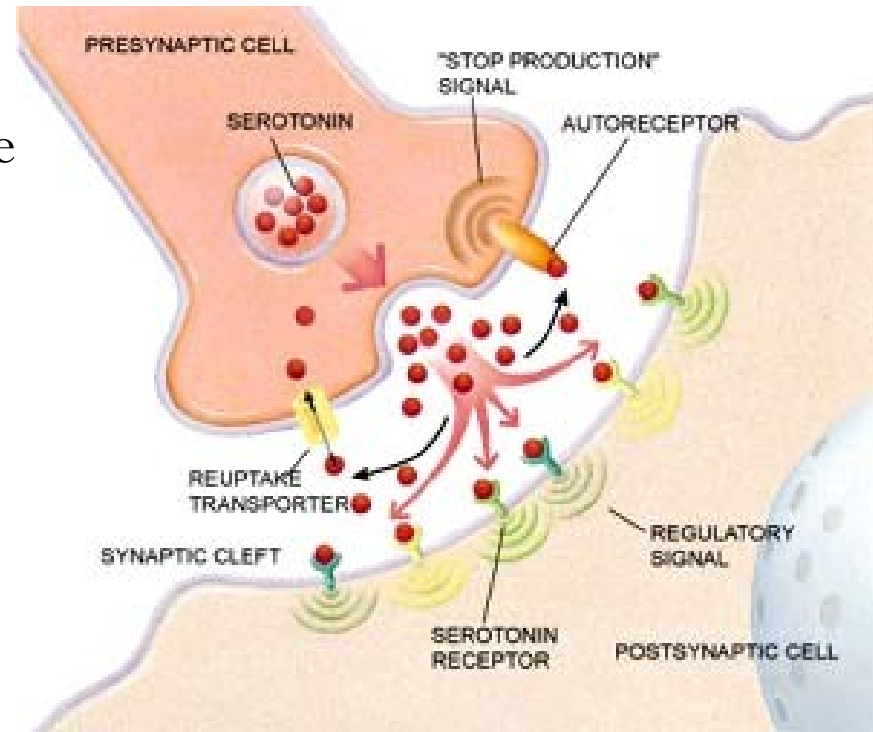
- Attributed to a deficiency in **neurotransmitter transmission** in the **CNS**
- Successful antidepressants affect a **combo of NE, 5-HT, and histamine reuptake**, but do only weakly act on DA

The Synapse



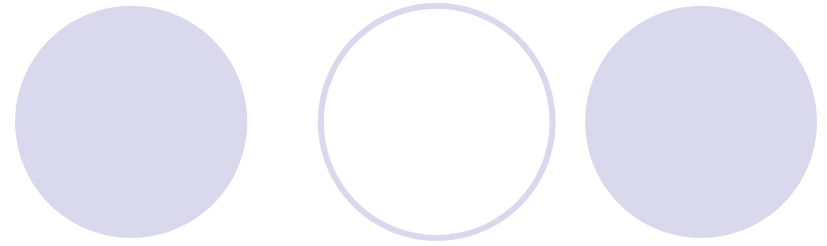
The serotonin transporter (SERT)

- Positioned periplanar to the pre-synaptic axon terminal
- **Role-** move serotonin back into the pre-synaptic cell for future use
- **Opposite activity of the serotonin receptors** that are positioned post-synaptically
- Fundamental in a biochemical understanding of depression



<http://www.psycheducation.org/mechanism/4WhyShortsLongs.htm>

The Antidepressants



- MAOIs

- Oldest antidepressant class that was discovered in 1952 with the use of iproniazid.

- TCAs

- SSRIs (1970s)

- Atypicals



Alternative therapies

- Electroconvulsive Therapy (ECT)

- Seizure induced using an electric current passed through the brain

- Exercise

- Counseling

- Lithium- commonly coupled with other treatments

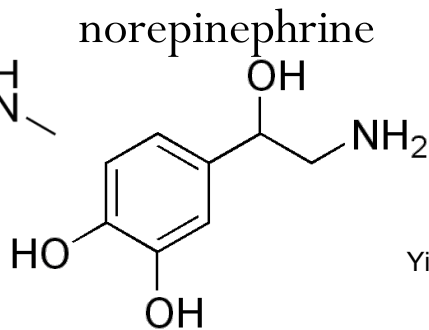
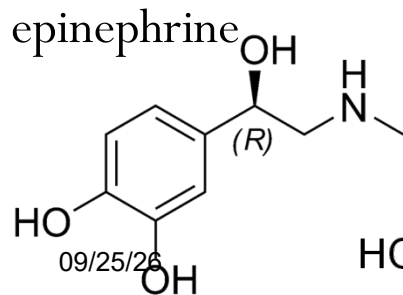
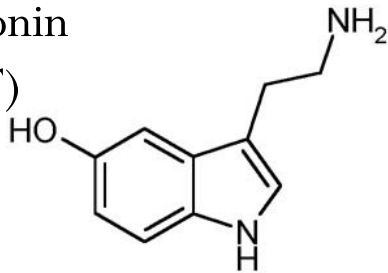
- Natural Supplements

09/25/26 ○ St. John's Wort

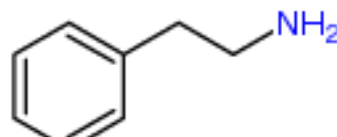
Monoamine Oxidase Inhibitors (MAOI)

- MAO's degrade amines in the nervous system
 - MAO-A deaminates serotonin, norepinephrine, and epinephrine
 - MAO-B degrades phenethylamine
- A and B types both degrade dopamine and tyramine

Serotonin
(5-HT)

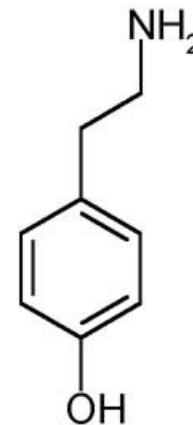


phenethylamine

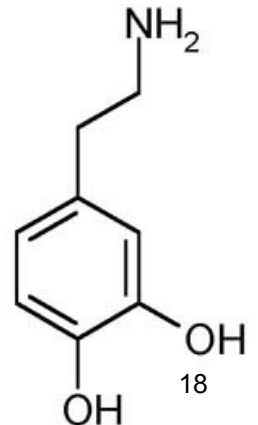


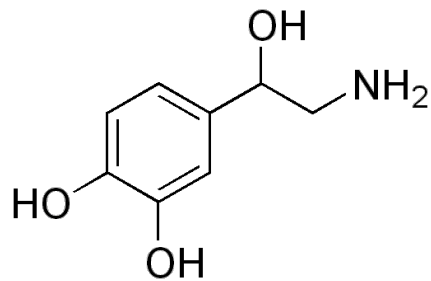
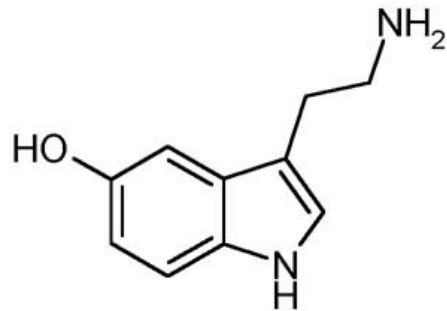
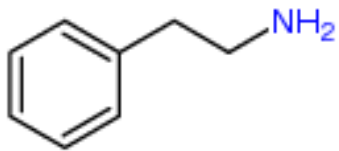
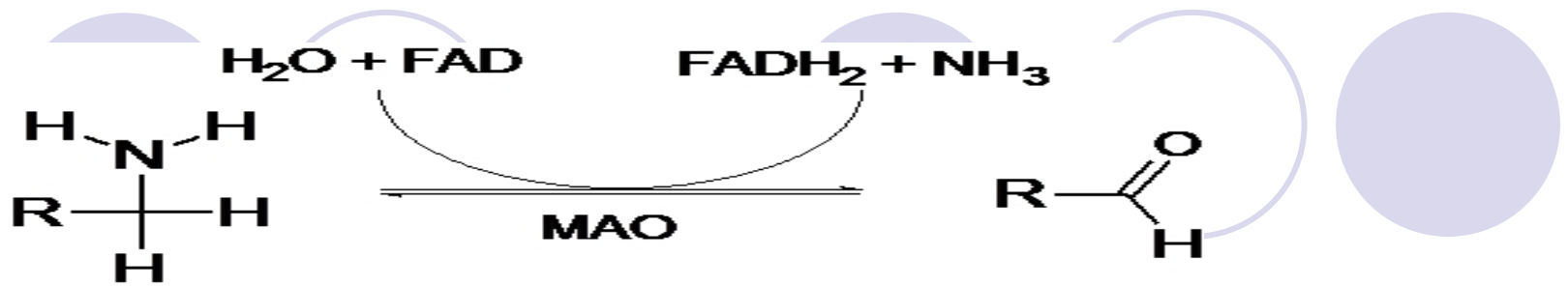
Yitagesu Tewabe (MSc.)

tyramine



dopamine

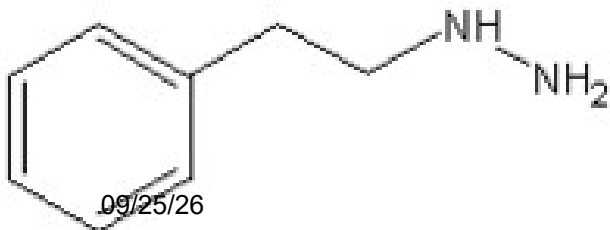




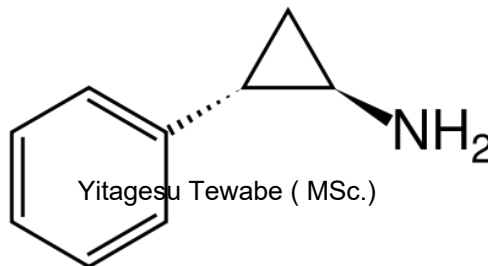
MAOI's continued...

- Thus, MAOI's cause an increase in the biogenic amine concentration
 - MAOI-A = clorgyline
 - MAOI-B = selegiline (approved for treatment of Parkinson's)
 - Nonselective MAOI = phenelzine, tranylcypromine, isocarboxazid
 - **MAOI-A's** are thought to be more effective in treating **major depression**

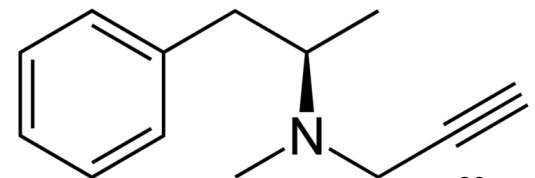
Phenelzine
(NARDIL)



Tranylcypromine
(PARNATE)



Selegiline
(ELDEPRYL)

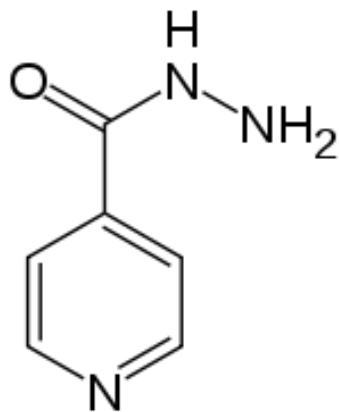


MONOAMINE OXIDASE (MAO) AND DEPRESSION

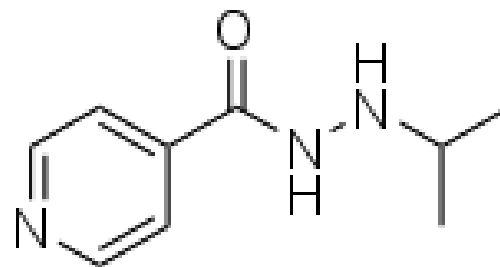
- MAO catalyze deamination of intracellular monoamines
 - MAO-A oxidizes epinephrine, norepinephrine, serotonin
 - MAO-B oxidizes phenylethylamine
 - Both oxidize dopamine nonpreferentially
- MAO transporters reuptake extracellular monoamine

MONOAMINE OXIDASE INHIBITORS (MAOIS)

- Iproniazid was resulted by isopropyl substitution of Isoniazid
 - **Iproniazid inhibits MAO**



Isoniazid



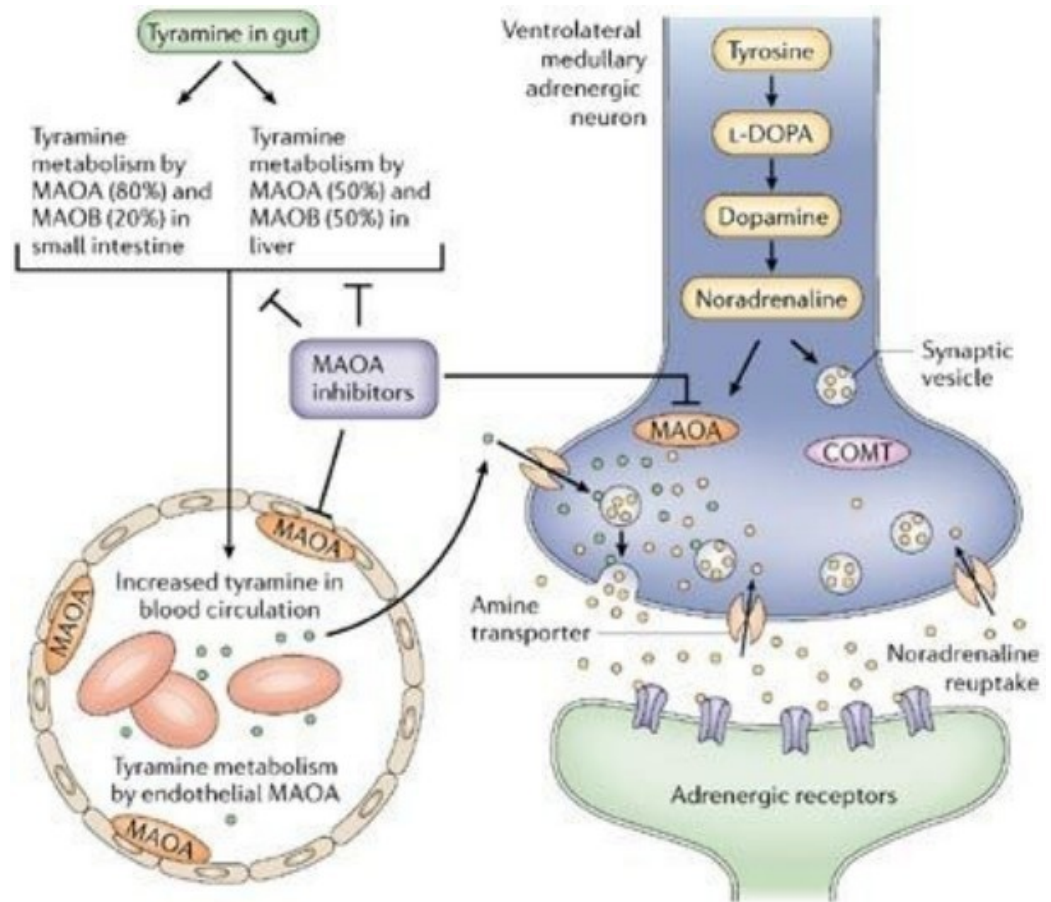
Iproniazid

MAOI ON THE MARKET

- MAO Inhibitors (nonselective)
 - Phenelzine (Nardil)
 - Tranylcypromine (Parnate)
 - Isocarboxazid (Marplan)
- MAO-B Inhibitors (selective for MAO-B)
 - Selegiline (Emsam)

MAOI MOA

- MAO contains a cysteinyl-linked flavin
- MAOIs covalently bind to N-5 of the flavin residue of the enzyme



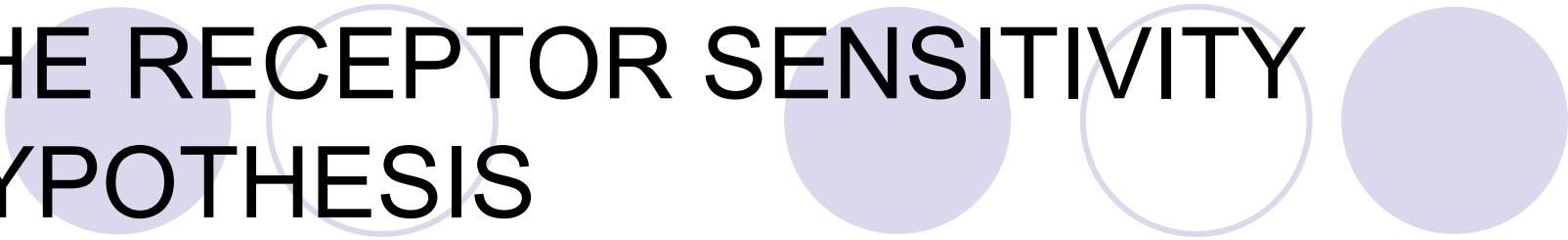
MAOI SIDE EFFECTS

- **Drowsiness/Fatigue**
- Constipation
- Nausea
- Diarrhea
- Dizziness
- **Low blood pressure**
- Lightheadedness,
- Decreased urine output
- **Decreased sexual function**
- Sleep disturbances
- Muscle twitching
- **Weight gain**
- Blurred vision
- Headache
- Increased appetite
- Restlessness
- Shakiness
- Weakness
- Increased sweating

MAOI SIDE EFFECTS

- Side effects have put MAOIs in the second or third line of defense despite superior efficacy
- MAO-A inhibitors interfere with breakdown of tyramine
 - High tyramine levels cause hypertensive crisis (the “cheese effect”)
 - Can be controlled with restricted diet
- MAOIs interact with certain drugs
 - Serotonin syndrome (muscle rigidity, fever, seizures)
 - Pain medications and SSRIs must be avoided

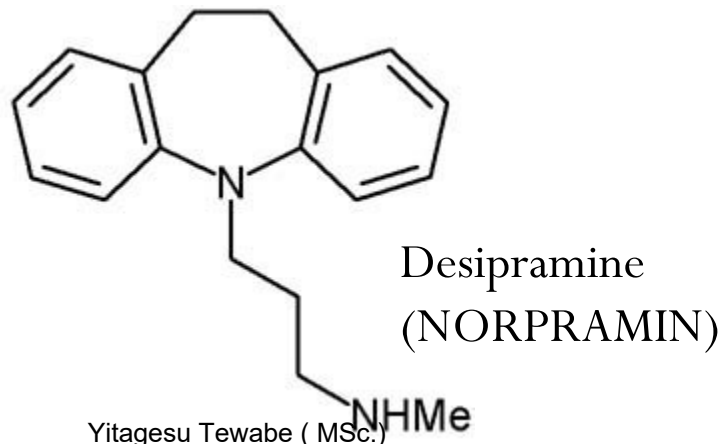
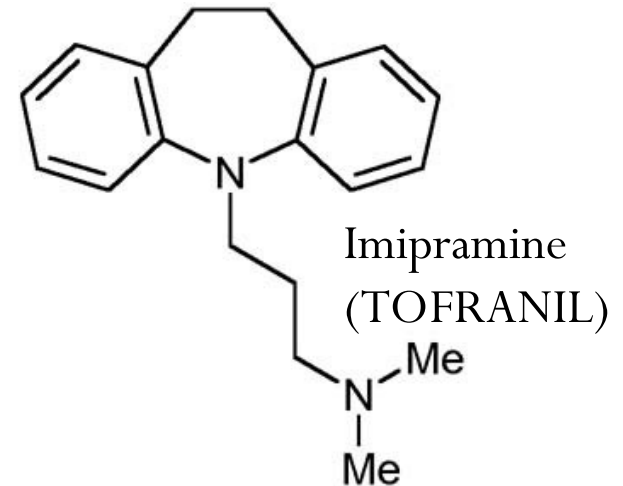
THE RECEPTOR SENSITIVITY HYPOTHESIS



- Supersensitivity and up-regulation of post-synaptic receptors leads to depression
- Suicidal and depressed patients have increased 5HT- α_2 receptors

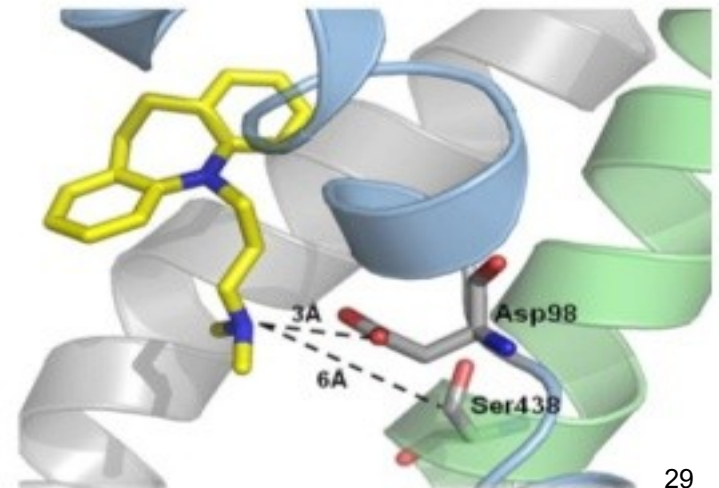
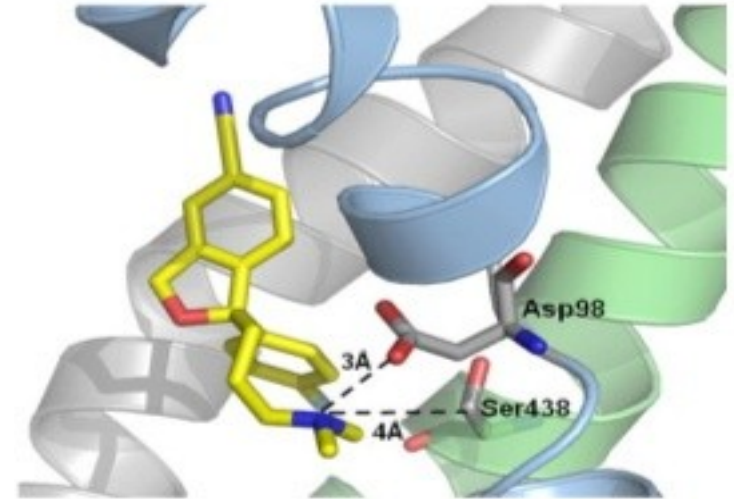
Tricyclic Antidepressants (TCA's)

- Imipramine discovered in 1958
 - Other TCA's are simply modifications of imipramine
- Work by inhibiting NE transport and variably inhibiting 5-HT transport

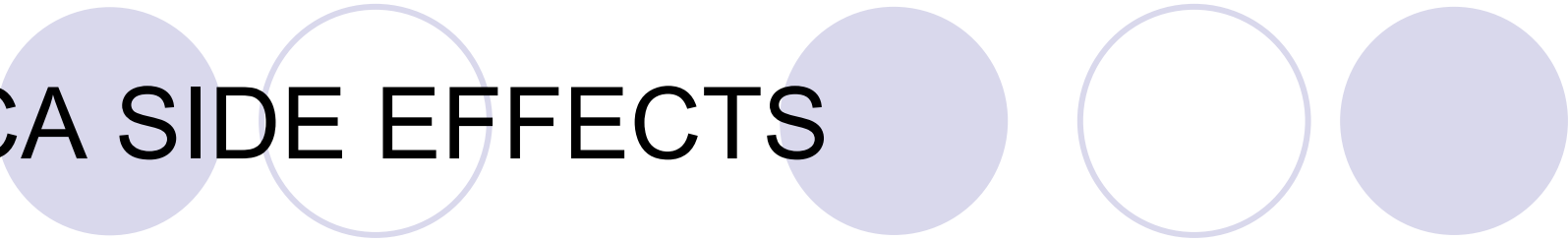


TCA MECHANISM OF ACTION

- TCAs inhibit serotonin, norepinephrine, and dopamine transporters, slowing reuptake
- TCAs also allow for the downregulation of post-synaptic receptors
- All TCAs and SSRIs contain an essential amino group that appears to interact with Asp-98 in hSERT

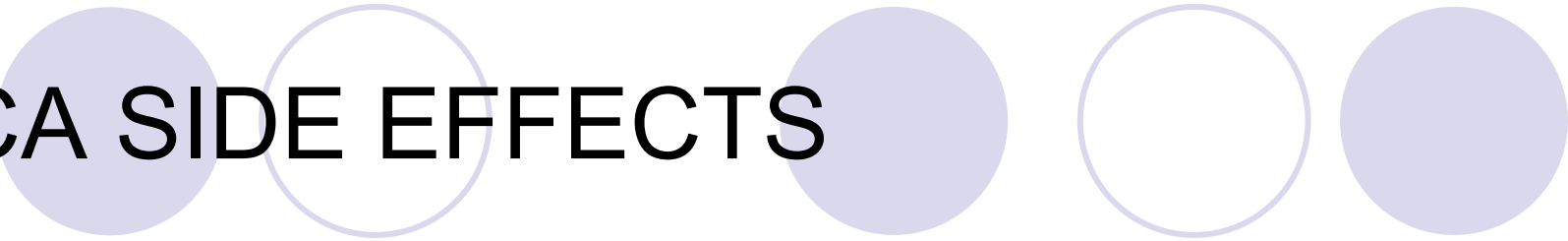


TCA SIDE EFFECTS



- Muscarinic M1 receptor antagonism - anticholinergic effects including dry mouth, blurred vision, constipation, urinary retention and impotence
- Histamine H1 receptor antagonism - sedation and weight gain
- Adrenergic α receptor antagonism - postural hypotension
- Direct membrane effects - reduced seizure threshold, arrhythmia
- Serotonin 5-HT₂ receptor antagonism - weight gain (and reduced anxiety)

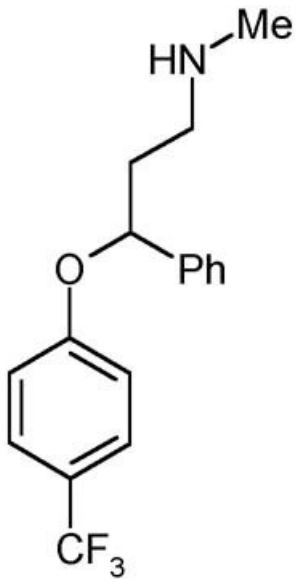
TCA SIDE EFFECTS



- Nonselectivity results in greater side effects
- TCAs can also lead to cardiotoxicity
 - Increased LDH leakage
 - Slow cardiac conduction
- High potency can lead to mania
 - Contraindicated with persons with bipolar disorder or manic depression

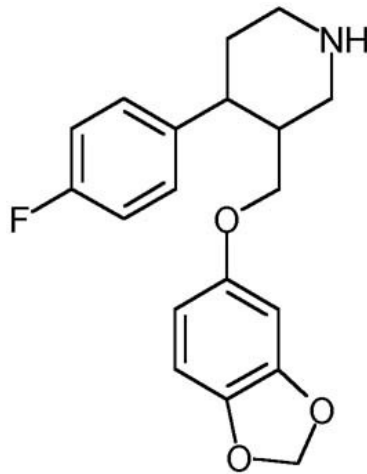
Selective Serotonin Reuptake Inhibitors

- Most prescribed type due to ↓ toxicity
- Treatment of choice for OCD

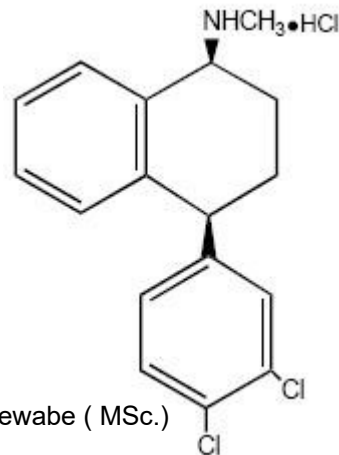


Fluoxetine
(PROZAC)

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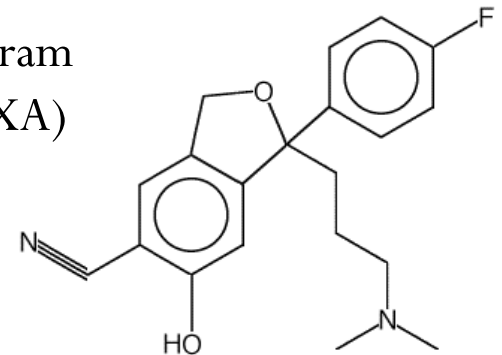


Paroxetine
(PAXIL)



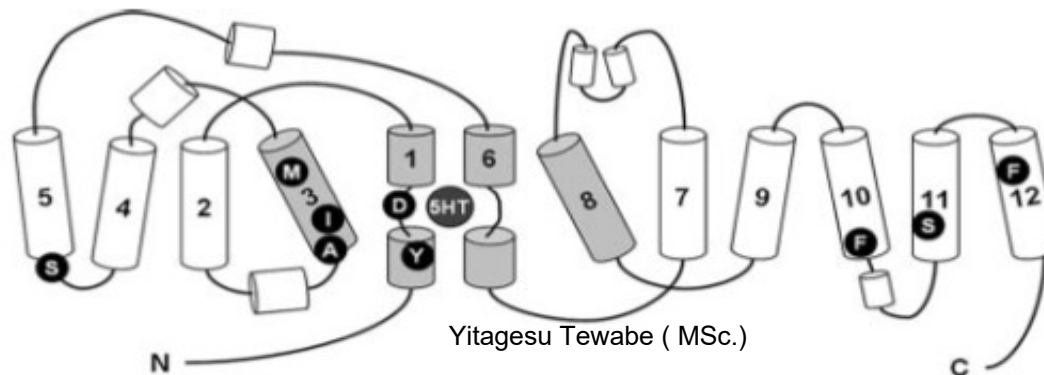
Sertraline
(ZOLOFT)

Citalopram
(CELEXA)



SSRIS MECHANISM OF ACTION

- Exact mechanism remains uncertain
- Ser-438 residue in the human serotonin transporter (hSERT) appears to be a determining factor in SSRI potency
- Antidepressants interact directly with hSERT



SEROTONIN-NOREPINEPHRINE REUPTAKE INHIBITORS (SNRIS)

- Slightly greater efficacy than SSRIs
- Slightly fewer adverse effects than SSRIs
- Current drugs

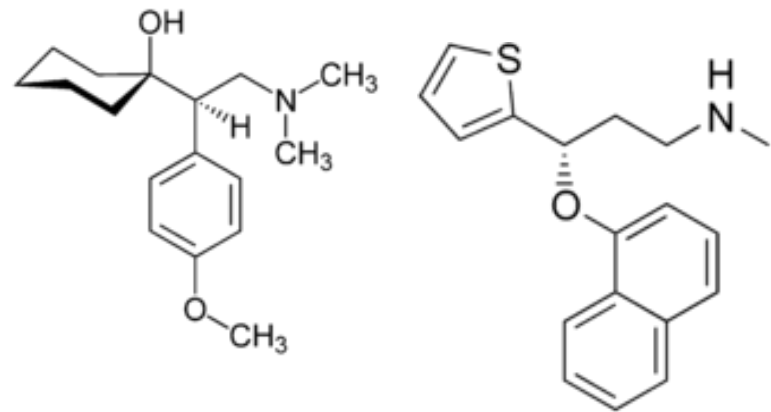
- Venlafaxine (Effexor)
- Duloxetine (Cymbalta)

- Mechanism of Action

- Very similar to SSRIs
- Works on both neurotransmitters

- Side effects

- Similar to SSRIs
- Suicide

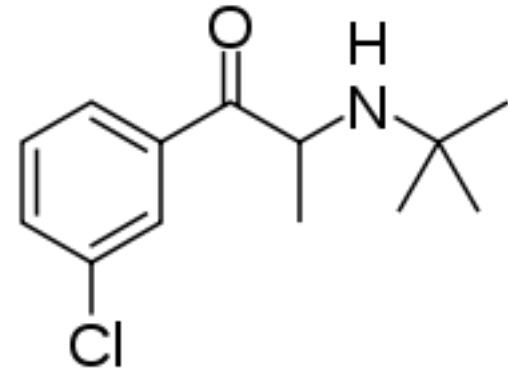


Venlafaxine Duloxetine

NOREPINEPHRINE-DOPAMINE REUPTAKE INHIBITORS (NDRIS)

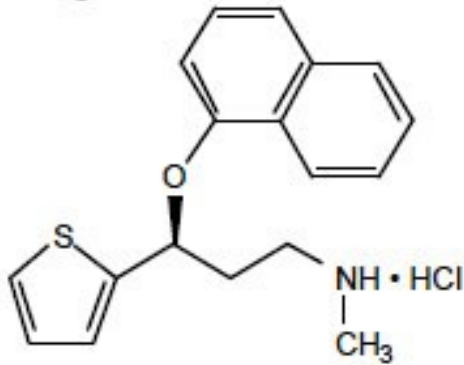
- Current drugs
 - Bupropion (Wellbutrin)
- Mechanisms of Action
 - Similar to SSRIs and SNRIs
 - More potent in inhibiting dopamine
 - Also an $\alpha 3$ - $\beta 4$ nicotinic antagonist

- Adverse effects
 - Lowers seizure threshold
 - Suicide
 - Does not cause weight gain or sexual dysfunction (even used to treat the two)



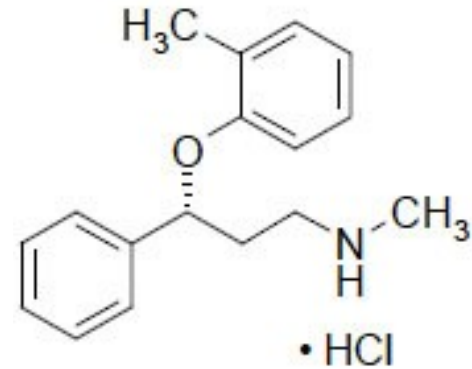
Bupropion

Atypical antidepressants



Duloxetine
(CYMBALTA)

Targets NE, 5-HT



Atomoxetine
(STRATTERA)

Targets NE

● As widely prescribed as SSRI's

Side effects

TCA's and MAOI's

- Generally, not prescribed unless patient does not respond to newer drugs
- Cardiotoxic at high doses
 - Lipophilic, strongly bind to tissues, affinity for cardiac
 - Many cardiac side effects!
- Dry mouth
- Constipation
- Dizziness
- Blurred vision
- Urinary retention

SSRI's and newer antidepressants

- Fewer side effects, less toxic
- Nausea/vomiting
- Headache
- Sexual dysfunction

Long-term effects on brain chemistry

↓ Tyrosine hydroxylase

↓ Norepinephrine transporter protein

↓ Postsynaptic beta adrenergic receptors

↓ GABA receptors

Altered sensitivity to muscarinic acetylcholinase
receptors

↑ cAMP production

QUESTIONS

1. Many of the medications to treat depression are thought to involve systems utilizing the monoamine neurotransmitters, noradrenaline, dopamine, and serotonin (5-HT). Draw the structures of these neurotransmitters. Why are they called monoamines? Illustrate their structural resemblance to one another.
2. Monoamine oxidase inhibitors (MAOIs) increase CNS synaptic concentrations of these monoamines by inhibiting an enzyme responsible for their degradation. Draw the reaction scheme for the biological degradation of noradrenaline by monoamine oxidase.
3. Illustrate how the TCAs and SSRIs might resemble the monoamine neurotransmitters, providing one example of each class of antidepressant.